

CariSurg MedTech Pathways 2026

Feasibility Memo: Yale EMLLC Triage Dataset

Student: Ashi Baldie

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Programme: CariSurg MedTech Pathways 2026

GitHub Repository

<https://github.com/baldiea75-tech/carisurg-portfolio.git>

Verdict

Proceed to a baseline model, with caveats. The dataset is suitable for a first triage model, provided the two outcome only columns are excluded from training and no result is applied to a Caribbean ED without local validation first.

Dataset Summary

- 55,121 emergency department visits, 225 pieces of information recorded per visit.
- Patients aged 18 to 107, average age 55.
- 57.6 percent Female, 42.4 percent Male.
- Race: 53.4 percent White, 29.0 percent Black or African American, 16.4 percent Other, Unknown or Refused.
- Every visit carries a real nurse assigned urgency score, ESI, where 1 is most urgent and 5 is least urgent. Most visits are mid urgency: 32.5 percent are ESI 2, 49.0 percent are ESI 3. Very few, 0.1 percent or 77 patients, are the most urgent category, ESI 1.
- Seven vital signs (heart rate, blood pressure, breathing rate, oxygen level, temperature, blood sugar) and 200 yes or no flags for the patient's presenting complaint (chest pain, abdominal pain and so on) are recorded for every visit.
- No gaps at all. Every one of the 225 fields is filled in for every one of the 55,121 visits, confirmed visually in `figures/missingness_heatmap.png`.
- Temperature is recorded in Fahrenheit, not Celsius, with an average of 98.09 degrees.

Top Concerns

1. ***Very few of the sickest patients:*** severe class imbalance. Only 77 of 55,121 visits, 0.1 percent, are the most urgent category. A model trained naively on this data could learn to almost never predict "most urgent" and still look accurate overall, which is exactly the failure mode this project cannot afford in triage. Mitigation: use stratified sampling or class weighting when training in Week 6, and judge the model on how well it catches this rare group specifically, not on overall accuracy.
2. ***Two columns leak the answer:*** outcome leakage. `disposition`, whether the patient was admitted or discharged, and `previousdispo` are only known after the visit ends. A nurse doing triage would never have this information at the moment of decision. Mitigation: both columns are excluded from every feature set used in modelling.
3. This is a US hospital's patients, not a Caribbean ED's. The dataset comes from a single academic hospital in the United States. The mix of patients, complaints and available resources at Mercer General will likely differ, possibly substantially. Mitigation: every finding in this memo is caveated as a US benchmark, not a proven result for Mercer. No model built

on this data should be deployed locally without a validation study on Mercer's own patients first.

Top Reasons to Proceed

1. A real triage label, not a stand in. Every visit carries a genuine nurse assigned urgency score with no gaps and a sensible spread across all five levels, so the real target can be learned rather than an approximation of it.
2. Clean, usable vital signs. After correcting 29 clinically impossible readings, all seven vital signs are complete and move in the clinically expected direction with urgency. For example, lower oxygen levels track with higher urgency, exactly as a clinician would expect.
3. Enough patients for an honest test. 55,121 visits is large enough to train a first model and still hold back a separate group of patients purely for testing, so any performance claim can be trusted rather than being an artefact of the training data.

Caveats

- Temperature is in Fahrenheit. Use 100.4 degrees as the fever threshold, not 38 Celsius.
- Most of the 200 complaint flags are rare. 174 of 200 appear in fewer than 1 percent of visits and carry little signal on their own; these will be dropped before modelling.
- The relationships found here are associations, not proof. A vital sign moving with urgency does not mean it causes that urgency. This is useful for ranking candidate features, not for clinical claims.
- This has not been tested on a Caribbean ED population. Treat every number in this memo as a US benchmark until validated locally.
- Two minor formatting quirks were found and are harmless but worth fixing before modelling: the urgency score and the 200 complaint flags are stored as decimal numbers, for example 1.0, rather than whole number categories. This has no effect on the analysis in this memo, but is worth tidying in Week 6.

6. Features Shortlisted for Week 6

Leakage columns are excluded from this shortlist on purpose.

Rank	Feature	Reason
1	age	Strongest single correlate (r = negative 0.24). Older patients tend to present more acutely unwell
2	triage_vital_o2 (oxygen level)	Directly signals respiratory failure. Strongest vital sign correlate (r = positive 0.18)
3	triage_vital_rr (breathing rate)	Early sign of a patient deteriorating (r = negative 0.10)
4	triage_vital_hr (heart rate)	Signals cardiovascular stress (r = negative 0.10)
5	triage_glucose (blood	Flags diabetic emergencies and sepsis (r = negative 0.08)

	sugar)	
6	trriage_vital_temp (temperature)	Flags infection and hypothermia
7	trriage_vital_sbp (systolic blood pressure)	Flags dangerously low or high blood pressure
8	cc_chestpain	Red flag complaint. Most strongly associated with high urgency of any complaint (r = negative 0.16)
9	cc_shortnessofbreath	Second most strongly associated red flag complaint (r = negative 0.15)
10	cc_alteredmentalstatus	Direct sign of a neurological emergency

disposition and previousdispo are excluded from this list and from all model training. See Concern 2 above.

Supporting Evidence

All figures referenced above are committed alongside this memo in docs/figures/:

- missingness_heatmap.png: confirms zero missing data across all structured columns
- plot1_esi_distribution.png: the urgency score class imbalance described in Concern 1
- plot2_demographics.png: the race and insurance breakdown behind the representativeness caveat
- plot3_vitals_distributions.png: the cleaned vital sign distributions behind Reason 2
- plot4_chief_complaints.png: the complaint frequency behind the sparsity caveat

Full methodology, code and the complete profiling output are in notebooks/Week5_Interim_Data_Exploration.ipynb.